

Bayesian quantile correction and synthesis for climate products

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QUESTION

Correction before synthesis

How can historical archives estimate source corrections before forecasts with different horizons are blended into one predictive distribution?

Case study: daily San Lorenzo River flow at the USGS Big Trees gauge, modeled on the $\log(1 + x)$ scale with x in m^3/s . Before each forecast origin, USGS observations and retrospective ECMWF/-GloFAS and NOAA/NWS products identify time-varying source corrections. After that origin, issued ensembles and exogenous information produce source-adjusted predictive flow quantiles.



USGS observations
15-min public gauge data; daily log-flow target for state updating and held-out verification.



ECMWF GloFAS
retrospective alignment; daily 51-member forecast guidance to 28 days.

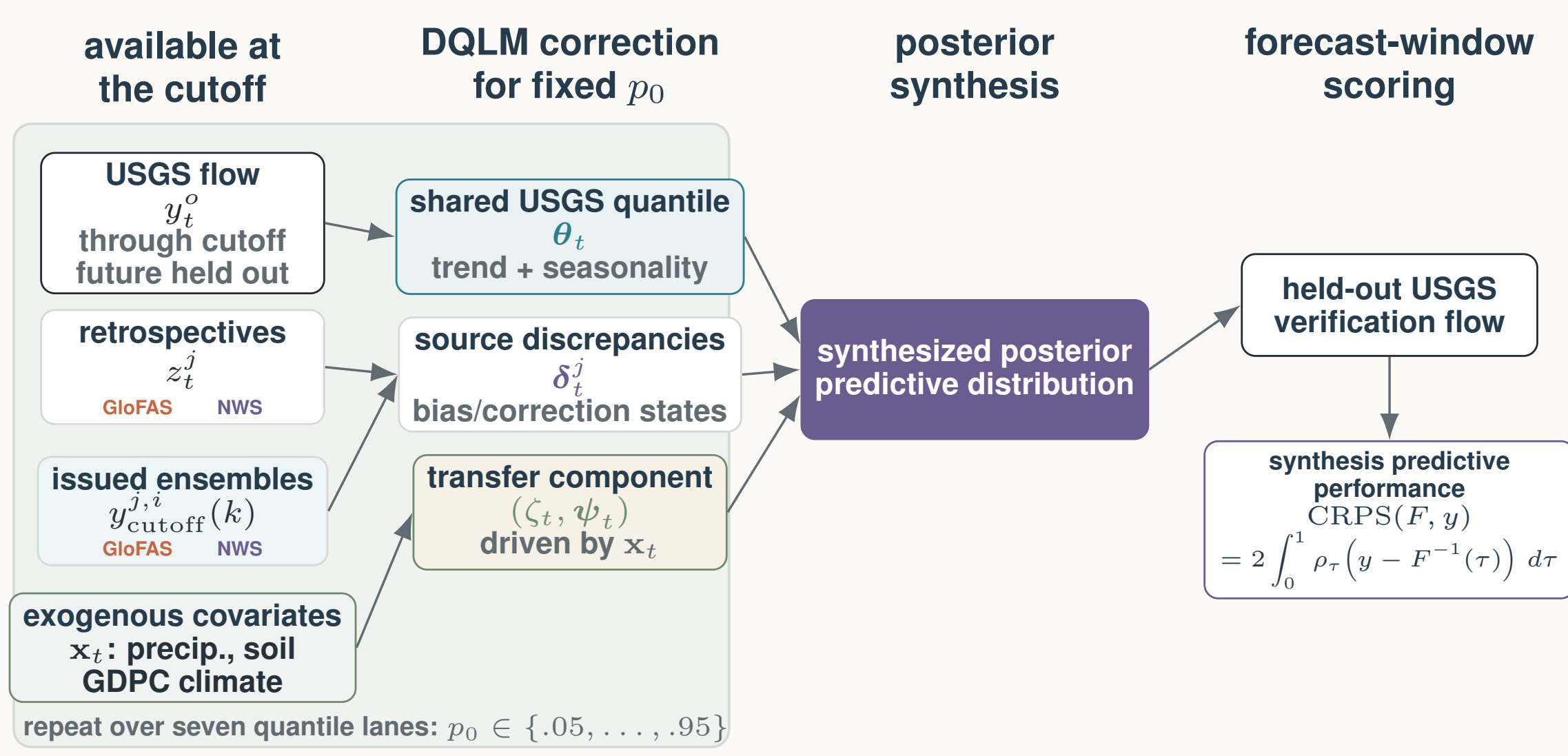


NOAA NWS
retrospective alignment; hourly forecasts evaluated on common daily leads 1–8.

Exogenous information
Local basin precipitation and soil moisture; the first generalized dynamic principal component (GDPC) summarizes key Pacific, Atlantic, and global climate indices.

MODEL

Main Workflow



Source-aware DQLM workflow. For each target quantile p_0 , pre-origin USGS observations and retrospective product histories identify a latent river-flow quantile and source-specific correction states. After the cutoff, issued ensemble forecasts and staged exogenous covariates propagate corrected forecast-window quantiles. The seven fitted quantile lanes are then synthesized into one posterior predictive distribution and scored only with held-out USGS observations.

Dynamic Quantile Linear Model

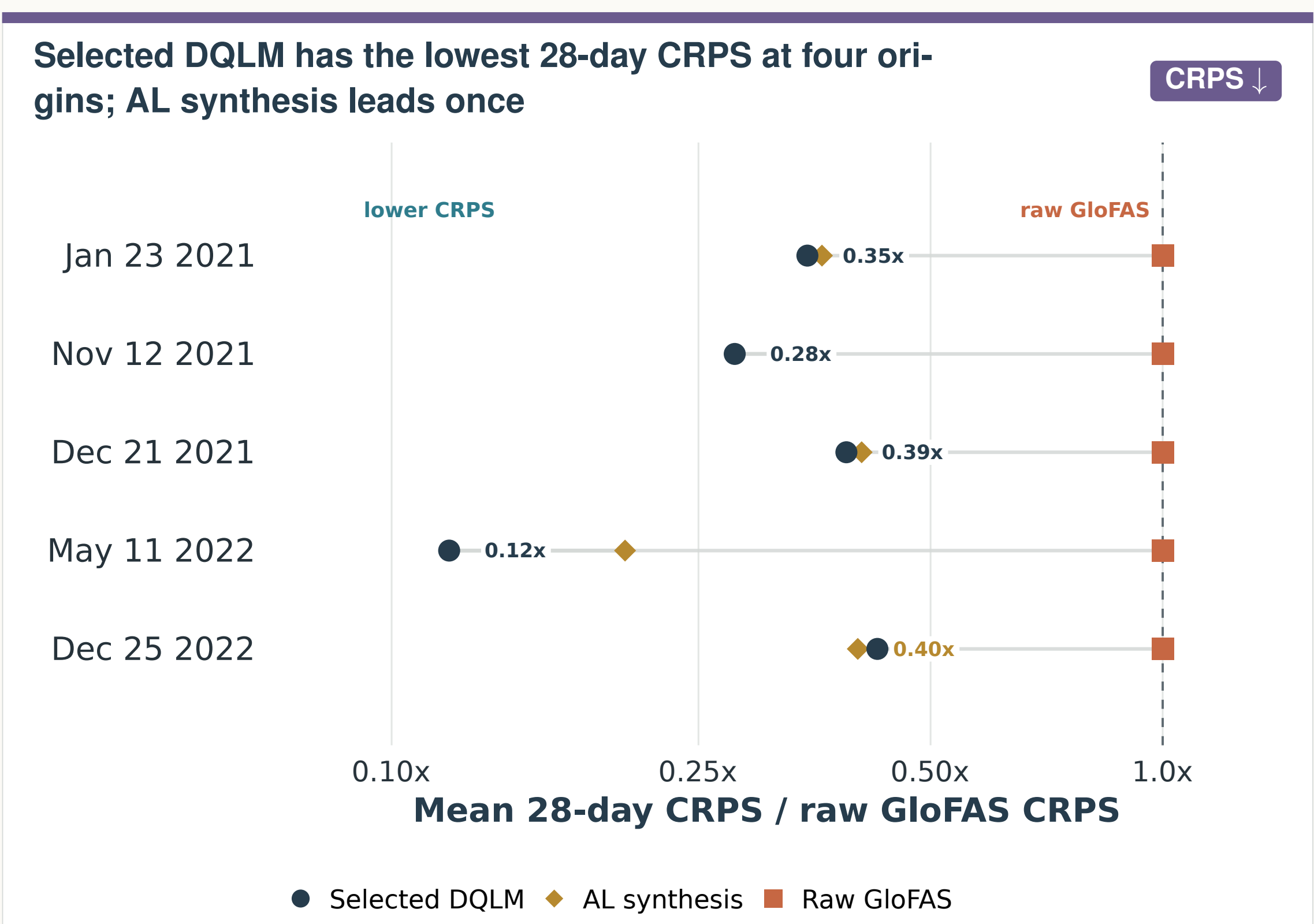
For each target quantile p_0 , let T denote the forecast origin/cutoff. The Dynamic Quantile Linear Model (DQLM) links observed flow, retrospective product histories, and issued forecast members through one latent river-flow quantile. Product-family discrepancies learn source corrections before the origin; the same source channels, with forecast-window covariates $\mathbf{x}_{T+k}$, then define corrected predictive quantiles after the origin.

$$\begin{aligned} \text{Observed measurements: } y_t^o &\sim \text{exAL}_{p_0}(\mathbf{F}_t^o \boldsymbol{\theta}_t + \zeta_t, \sigma^o, \gamma^o), & 1 \leq t \leq T, \\ \text{Retrospective products: } z_t^j &\sim \text{exAL}_{p_0}(\mathbf{F}_t^j(\boldsymbol{\theta}_t + \boldsymbol{\delta}_t^j) + \zeta_t, \sigma^j, \gamma^j), & j \in \mathcal{J}, 1 \leq t \leq T, \\ \text{Forecast products: } y_{\text{cutoff}}^{j,k}(k) &\sim \text{exAL}_{p_0}(\mathbf{F}_{T+k}^j(\boldsymbol{\theta}_{T+k} + \boldsymbol{\delta}_{T+k}^j + \zeta_{T+k}, \sigma^j, \gamma^j), & j \in \mathcal{J}, i \in \mathcal{I}_j, k \in \mathcal{K}_j, \\ \text{Trend and seasonal states: } \boldsymbol{\theta}_t | \boldsymbol{\theta}_{t-1} &\sim \mathcal{N}(\mathbf{G}_t \boldsymbol{\theta}_{t-1}, \mathbf{W}_t^{\boldsymbol{\theta}}), & 1 \leq t \leq T + K_{\max}, \\ \text{Source correction states: } \boldsymbol{\delta}_t^j | \boldsymbol{\delta}_{t-1}^j &\sim \mathcal{N}(\mathbf{G}_t^j \boldsymbol{\delta}_{t-1}^j, \mathbf{W}_t^{j,\boldsymbol{\delta}}), & j \in \mathcal{J}, 1 \leq t \leq T + K_{\max}, \\ \text{Exogenous adjustment: } \zeta_t | \zeta_{t-1}, \boldsymbol{\psi}_{t-1} &\sim \mathcal{N}(\lambda \zeta_{t-1} + \mathbf{x}_t^* \boldsymbol{\psi}_{t-1}, w_\zeta^2), & 1 \leq t \leq T + K_{\max}, \\ \text{Covariate-adjustment states: } \boldsymbol{\psi}_t | \boldsymbol{\psi}_{t-1} &\sim \mathcal{N}(\boldsymbol{\psi}_{t-1}, \mathbf{W}_t^{\boldsymbol{\psi}}), & 1 \leq t \leq T + K_{\max}. \end{aligned}$$

exAL_{p_0} is the extended asymmetric Laplace working likelihood for target quantile p_0 ; σ and γ are source-specific scale and skewness terms. T is the forecast origin, j a product family, i an ensemble member, k a lead, and $K_{\max}$ the longest horizon. The shared quantile state is $\boldsymbol{\theta}_t$; source discrepancies/corrections are $\boldsymbol{\delta}_t^j$; and $(\zeta_t, \boldsymbol{\psi}_t)$ is the exogenous-adjustment block driven by covariates $\mathbf{x}_t$, such as precipitation, soil moisture, and the GDPC climate-index summary. Component discounting controls the evolution variances.

RESULTS

28-day CRPS comparison



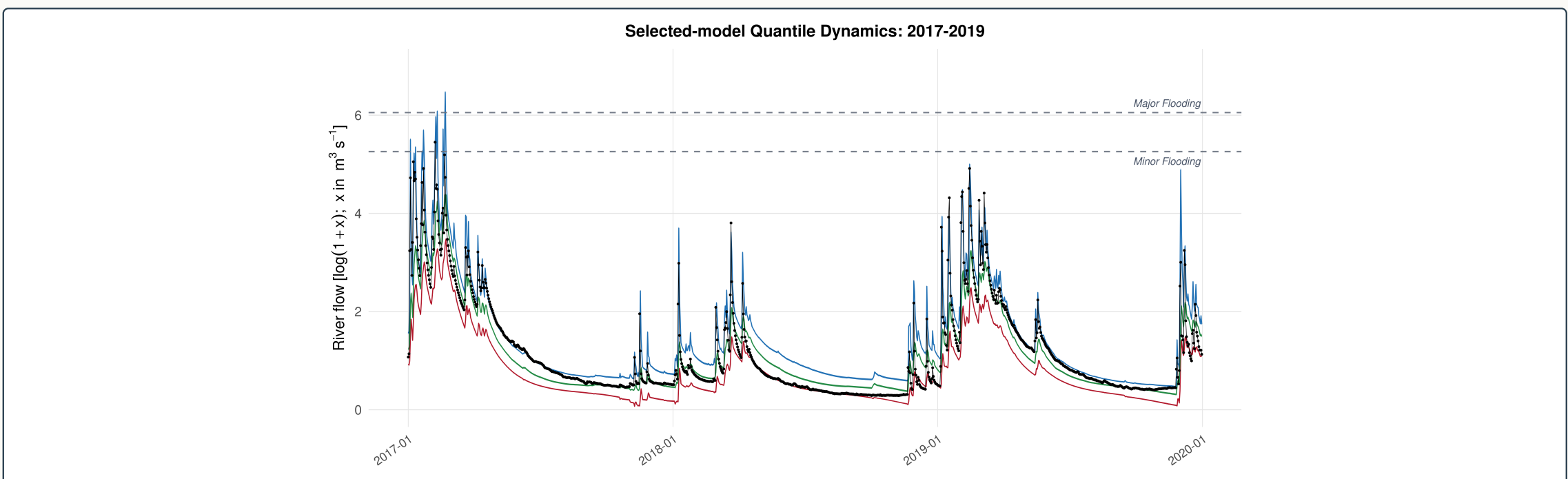
Exception: AL synthesis has the lowest 28-day CRPS at the 2022-12-25 origin.

How to read this panel. Each point is mean forecast-window CRPS divided by raw GloFAS CRPS at the same origin. Values below 1.0 improve on the uncorrected 28-day GloFAS benchmark.

Values left of the raw GloFAS line indicate lower mean CRPS than raw GloFAS at that origin; raw NWS is evaluated over common leads 1–8 in the horizon-matched tables, not mixed into this 28-day GloFAS-supported benchmark.

FIT

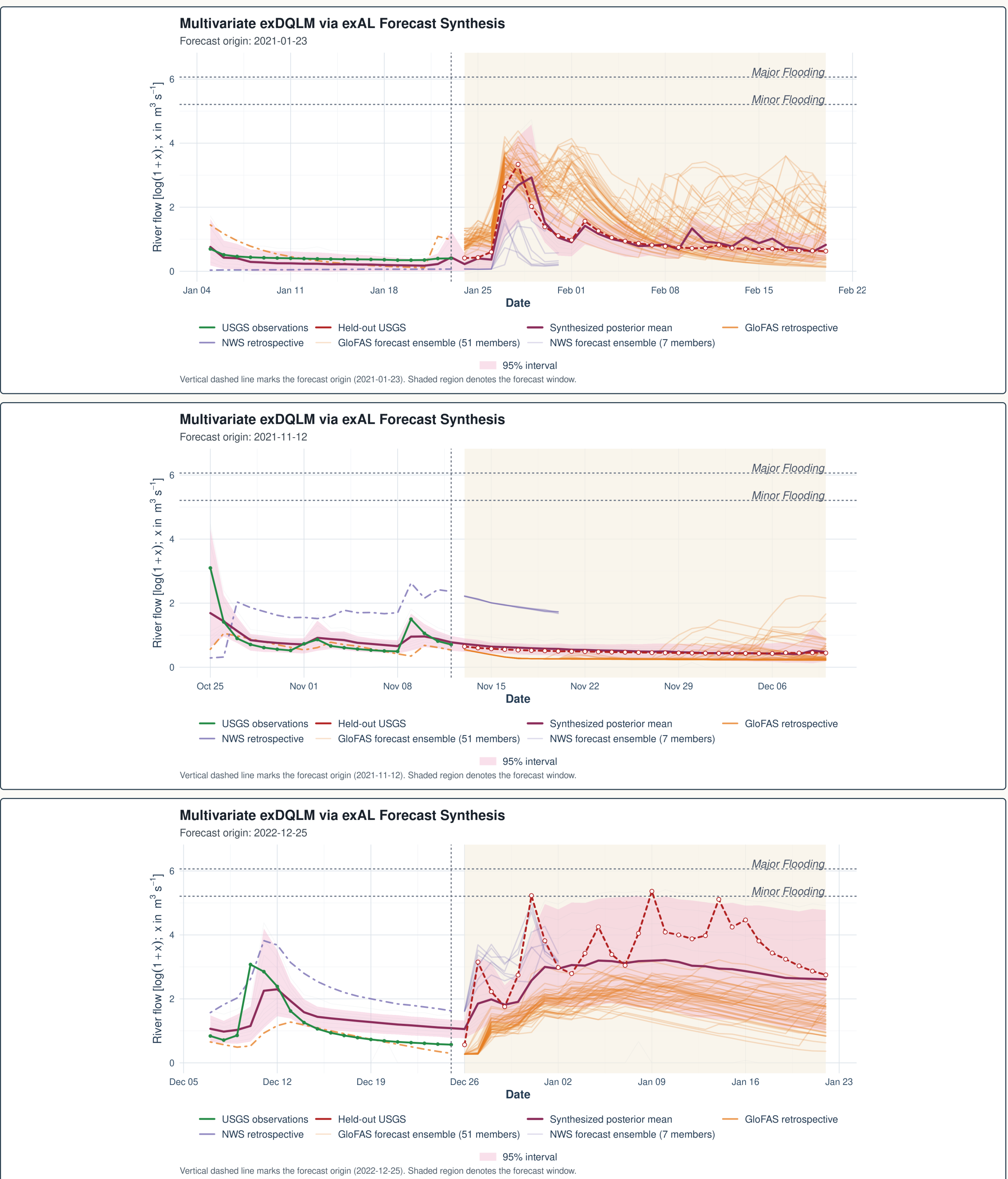
Wet-period quantile fit



Historical fit diagnostic. Selected-model fitted quantile locations during the 2017–2019 wet period. The 5th, 50th, and 95th quantile fits are shown with observed USGS flow; wider upper-tail bands make winter high-flow behavior visible.

EXAMPLE

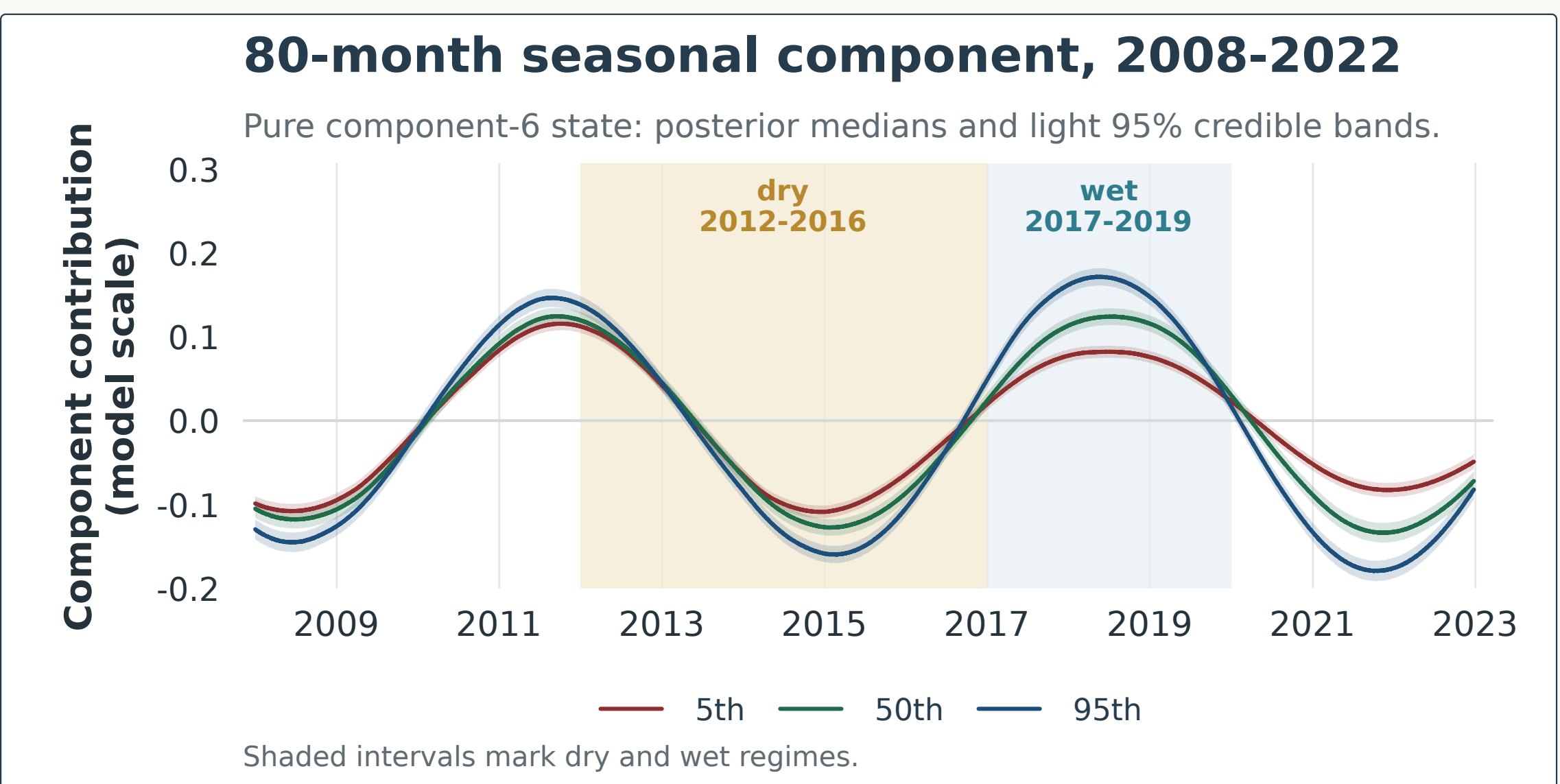
Three forecast-window examples



Origin-level illustrations. Three selected forecast windows show how corrected quantile forecasts combine into a posterior predictive envelope after the cutoff. The panels are illustrative checks of behavior at individual origins; the aggregate comparison remains the five-origin CRPS benchmark.

DYNAMICS

80-month seasonal component



Retrospective diagnostic. From the selected model at the 2022-12-25 origin. Lines show the isolated 80-month state component, without the trend component added, for the 5th, 50th, and 95th target-quantile lanes over 2008–2022; light ribbons show 95% credible bands and shaded intervals mark dry and wet regimes.

TAKEAWAYS

Main takeaways

- Historical archives learn source-specific corrections before forecast products are blended.
- Seven corrected quantile lanes are synthesized into one posterior predictive distribution.
- In this five-origin study, the selected DQLM has the lowest 28-day CRPS at four origins; 2022-12-25 favors AL synthesis.
- Fitted quantile dynamics remain interpretable: spread widens during wet high-flow periods.

